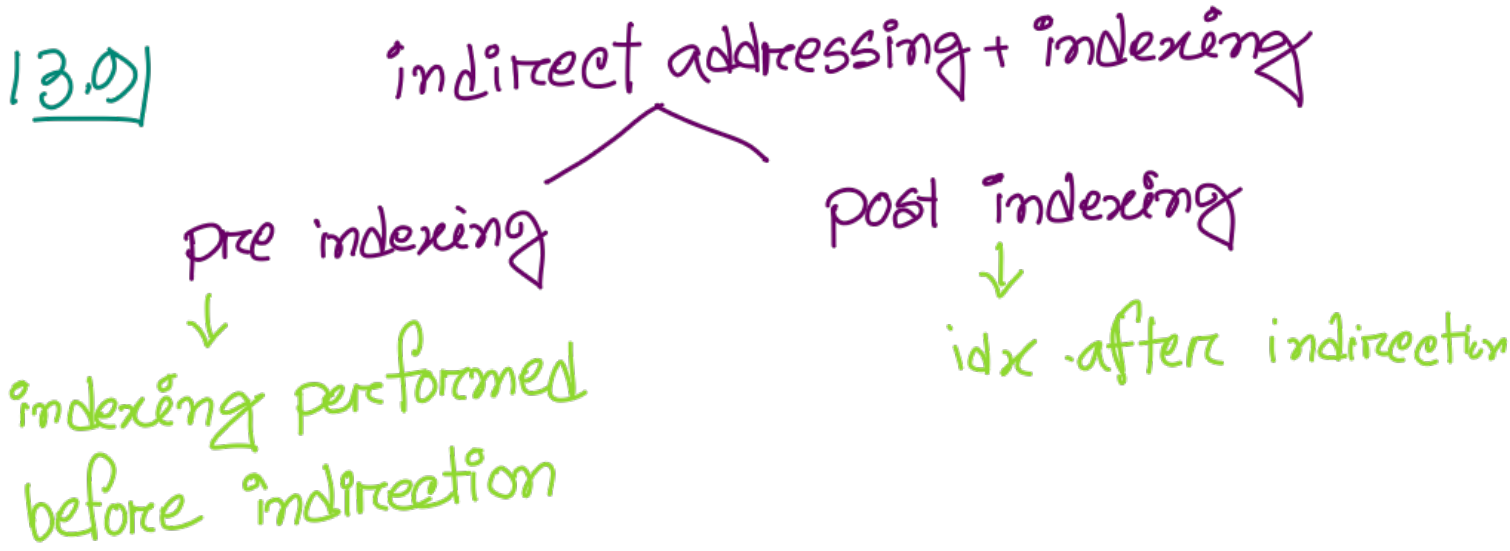
13.7| Relative addressing: The current instr's address is added to the address field to produce the EA.

13.8| auto incre, decre the idx reg after each reference.

13.9|



10| # of Addressing Modes.
" operand
reg vs memory
of reg sets.
Address range
" granularity.

11| variable length:
fixed: ... with diff

adv : large ... opcode length

More flexible addressing → various comb^n of reg + mem ref + addrsing mode.

disadv : Increase in complexity of CPU.

13.1 Given the following memory values and aone-address machine with an accumulator,what values do the following instructions load into the accumulator?■ Word 20 contains 40.■ Word 30 contains 50.■ Word 40 contains 60.■ Word 50 contains 70.

a. b. c. d. e. f. LOAD IMMEDIATE 20LOAD DIRECT 20LOAD INDIRECT 20LOAD IMMEDIATE 30LOAD DIRECT 30LOAD INDIRECT 30

a) Load imm. 20 → 20

b) " Direct " → 40

'm " 20 → 60

imm 30 → 30

Direct 30 → 50

'm " 30 → 70

13.2 Let the address stored in the program counter be designated by the symbol X1. Theinstruction stored in X1 has an address part (operand reference) X2. The operandneeded to execute the instruction is stored in the

memory word with address X3. Anindex register contains the value X4. What is the relationship between these variousquantities if the addressing mode of the instruction is (a) direct; (b) indirect; (c) PCrelative; (d) indexed?

⇒ a) Direct: $x3 = x2$ → instruction's address field.

b) indirect: $x3 = (x2)$ → content at x2

c) pc relative: $x_3 = x_2 + x_1 + 1$

d) indexed : $x_3 = x_4 + x_2$

14

An address field in an instruction contains decimal value 14. Where is the corresponding operand located fora. immediate addressing?b. direct addressing?c. indirect addressing?d. register addressing?e. register indirect addressing?

a) imm → The addrs field

b) direct → mem loc 14

c) in " → " " whose address is in mem loc 14

d) reg " → reg 14

e) reg indirect → mem loc whose address is in reg 14.

4 imm → 201 500

direct →

⑤ $(PC+3) + offset = Target$

$(256028+3) + (-31) = Target = 256000$

⑥ $(PC+1) + offset = Target$

$(620+1) + offset = 530$

$\therefore offset = -91$

$91 \xrightarrow{\text{in 10 bits}} 0001011011 = 1110100100 =$